

Planetary Science & Solar System Dynamics

Problem Set — Solutions

fizicyst

Section A — Objective

O1. Answer: (b). At 2 AU, $S = 1361/2^2 = 340.3 \text{ W/m}^2$. Then

$$T_{\text{eff}} = \left[\frac{S(1 - A)}{4\sigma} \right]^{1/4} = \left[\frac{340.3(0.70)}{4(5.67 \times 10^{-8})} \right]^{1/4} \approx (1.05 \times 10^9)^{1/4} \approx 180 \text{ K}.$$

O2. Answer: (b). $T_s = T_{\text{eff}}(n + 1)^{1/4} = 255 \times 4^{1/4} = 255 \times 1.414 \approx 361 \text{ K}$. (Option (c), 510 K, is the trap: multiplying by $(n + 1)$ rather than $(n + 1)^{1/4}$.)

O3. Answer: (c). $1/S = |1/1.00 - 1/1.88| = 0.4681 \implies S \approx 2.14 \text{ yr}$.

O4. Answer: (d) is false. Opposition requires the planet to sit opposite the Sun as seen from Earth, which is geometrically impossible for a planet whose orbit lies entirely inside Earth's — inferior planets reach inferior and superior conjunction instead. (a), (b), (c) are all correct.

O5. Answer: (b).

$$L_u = \frac{dR_p}{R_\odot - R_p} = \frac{(1.496 \times 10^{11})(1.737 \times 10^6)}{6.96 \times 10^8 - 1.737 \times 10^6} \approx 3.74 \times 10^8 \text{ m} \approx 3.7 \times 10^5 \text{ km},$$

slightly *shorter* than the mean Earth–Moon distance of $3.84 \times 10^5 \text{ km}$ — which is precisely why annular eclipses happen.

Section B — Subjective

S1.

(i) $a_t = \frac{r_1 + r_2}{2} = \frac{1.000 + 1.524}{2} = 1.262 \text{ AU}$. In solar units $t_{\text{transfer}} = \frac{1}{2}a_t^{3/2} = \frac{1}{2}(1.262)^{3/2} \approx 0.709 \text{ yr} \approx 259 \text{ days}$.

(ii) Departure ($r = 1.000 \text{ AU}$):

$$v_{c,1} = 2\pi = 6.283 \text{ AU/yr} = 29.78 \text{ km/s}, \quad v_p = \sqrt{4\pi^2 \left(2 - \frac{1}{1.262}\right)} = 6.905 \text{ AU/yr} = 32.73 \text{ km/s},$$

so $\Delta v_1 \approx 2.95 \text{ km/s}$. Arrival ($r = 1.524 \text{ AU}$):

$$v_{c,2} = 5.090 \text{ AU/yr} = 24.12 \text{ km/s}, \quad v_a = \sqrt{4\pi^2 \left(\frac{2}{1.524} - \frac{1}{1.262}\right)} = 4.531 \text{ AU/yr} = 21.47 \text{ km/s},$$

so $\Delta v_2 \approx 2.65 \text{ km/s}$, and $\Delta v_{\text{total}} \approx 5.60 \text{ km/s}$.

- (iii) $\gamma_0 = 180^\circ - \frac{360^\circ}{1.881}(0.709) = 180^\circ - 135.7^\circ \approx 44.3^\circ$ (Mars must lead Earth by this angle at launch). Synodic period: $1/S = |1/1.000 - 1/1.881| = 0.4684 \implies S \approx 2.14$ yr between windows.

S2.

- (i) $\tan(\delta/2) = \frac{GM_p}{bv_\infty^2} = \frac{1.267 \times 10^{17}}{(1.00 \times 10^9)(1.00 \times 10^4)^2} = 1.267$, so $\delta/2 = 51.7^\circ$ and $\delta \approx 103.4^\circ$.
- (ii) $\Delta v_{\text{helio}} = 2v_\infty \sin(\delta/2) = 2(10.0)(0.785) \approx 15.7$ km/s. Note this exceeds v_∞ itself by over 50% — the absolute ceiling is $2v_\infty = 20$ km/s, reached only for a 180° turn.
- (iii) Momentum conservation gives $m \Delta v_{\text{sc}} = M_p \Delta v_p$, so

$$\Delta v_p = \frac{(1000)(1.57 \times 10^4)}{1.898 \times 10^{27}} \approx 8 \times 10^{-21} \text{ m/s.}$$

Utterly unmeasurable — around 10^{-20} m/s against an orbital speed of 13 km/s. The spacecraft genuinely takes its energy from Jupiter's orbit, but the mass ratio makes the planet's share of the exchange completely negligible.

S3.

- (i) $S = 1361/(0.723)^2 \approx 2604$ W/m². Then

$$T_{\text{eff}} = \left[\frac{2604(1 - 0.77)}{4(5.67 \times 10^{-8})} \right]^{1/4} \approx (2.64 \times 10^9)^{1/4} \approx 227 \text{ K.}$$

- (ii) $n + 1 = \left(\frac{737}{227} \right)^4 = (3.25)^4 \approx 112$, so $n \approx 111$ layers.
- (iii) Venus intercepts about $1.9 \times$ Earth's flux, but reflects 77% of it, absorbing only $(1 - A) = 0.23$ against Earth's ≈ 0.70 . The net absorbed flux is therefore $1.9 \times 0.23/0.70 \approx 0.63$ of Earth's, and since $T \propto (\text{absorbed flux})^{1/4}$, Venus comes out at $(0.63)^{1/4} \approx 0.89$ of Earth's 255 K — about 227 K. High albedo beats high insolation.

The $n \approx 111$ should *not* be read as 111 physical layers of atmosphere. The layer model is a crude device in which each layer is idealised as perfectly transparent in the visible and perfectly opaque in the infrared; real atmospheres are neither, and absorb over a continuum of optical depths rather than in discrete slabs. What n actually encodes is roughly the infrared optical depth of Venus's CO₂ atmosphere — a measure of how many times outgoing radiation is absorbed and re-emitted before escaping. The large value correctly signals an extremely optically thick atmosphere; it carries no information about physical stratification.